

Detection is solved. Attention is not.

Event triage and coaching loop — a product design exercise for CameraMatics

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Every number in this deck is either measured with a named source, or an entry in an assumption register with a confidence and a way to test it. Nothing in between.

A triage and coaching loop, not an event detector

What it does

Every detected event is graded into one of four bands, each with its own action, and classified separately as risk, neutral or credit. The portal presents a weekly set that starts full and can be finished.

Why it is the right thing to build

A 100-vehicle fleet produces 50–450 detectable events a day. Most are technically correct and practically irrelevant. The manager reviews six of three hundred, then stops logging in. That is where every camera rollout dies.

The line the design turns on

Praise the good, coach the bad. A system that only ever detects failure makes invisibility the best outcome a driver can achieve — which is how a safety tool becomes the thing the cab resents.

50–450 events a day, one fleet, one manager, and no way to tell which six matter.

Range, not a point estimate — A-02, low confidence, and the assumption whose falsity would end this.

The deck is the submission. The repository is the evidence.

1

The product brain

Public repo — every assumption, decision, story, metric and review

github.com/dannydaley76/cameramatics-product-brain

2

The prototype

Low fidelity, real behaviour. Work through the group of five events

prototype/index.html

3

The story map

The whole capability on one page, sliced by release

storymap/index.html

About four hours of focused effort, spread across a week alongside other work. The volume is agent output under direction. The commit history shows both halves honestly — the elapsed days, and the bursts inside them that nobody produces by hand.

Six stages, and what each one had to produce

1 Discovery

Annotated brief, company analysis,
25 assumptions, 10 open questions

2 Solution design

14 decision records, each with the
alternatives it beat

3 Requirements

PRD, 4 epics, 15 stories with
acceptance criteria, a working prototype

4 Challenge

3 adversarial passes by a fresh reader.
All three returned “Not yet”

5 Delivery planning

Phasing against the dependency chain,
and a story map

6 Validation

17 metrics with counter-metrics, and
what would tell me I was wrong

What the agents did, and what they could not decide

What they did

- Generated the option sets I chose between
- Drafted every document in the repository
- Held roughly fifty cross-referenced IDs consistent
- Enforced the schema — the validator is committed, not described
- Attacked the work, as a separate reviewer with no memory of writing it
- Built the prototype and the 43 tests that check it

What they did not

- Choose the spine, or the primary user
- Decide what was cut from v1
- Set what counts as high severity — an ethical call before a modelling one
- Accept a risk on my behalf
- Close an open question
- Put a model anywhere in the product a rule would do

One honest caveat. Time constraints meant the agents did more of the primary research than I would normally allow — the usual shape is the reverse, where I draw the conclusions and feed those to the agent as context. Where it inverted here, the output is registered as an assumption rather than a finding, which is why six of the twenty-five assumptions exist at all.

Note the asymmetry, because it is deliberate: heavy agent use in the process, and a rule in canon that nothing in the product gets a model because a model was available.

Three times I overruled it

The judgement log records the decisions where an agent had a position worth disagreeing with — 27 entries, 8 of them overrules, 1 still unratified. It does not record the steering: the framing that happened in conversation before anything was drafted, which is the larger part and the part that decided what the agent produced at all. What follows is the checkable subset.

J-18

Praise the good, coach the bad

The agent had built a system that would coach a driver for braking to avoid a child. A detector that only finds failure makes invisibility the best a driver can achieve. Recognition is now half the product.

J-17

The number it invented

A fifteen-minute attention budget, cited across four documents until it looked sourced, hardened into a design target. I asked where it came from. It came from nowhere, and the register now records the withdrawal.

J-27

The lever it reached for

It measured a real asymmetry and proposed adding friction. Neutral means “nothing to see here” — charging someone to type that buys compliance text, not judgement. The fix was elsewhere.

A frame it missed, a fact it invented, a lever it reached for. Each entry names the class of error, because the pattern is more useful than the instance.

CHALLENGE

Then I had a fresh agent try to destroy it. Three times.

Requirements and decisions

18 attacks, 16 survived

NOT YET

The prototype

20 findings, driven in a browser

NOT YET

The three newest decisions

15 findings, 1 signed off

NOT YET

The finding worth reading

My prototype was fabricating. Confirming that a driver had avoided harm wrote a hardcoded sentence onto their record — the wrong day, the wrong road, the wrong act — under a named manager's confirmation, with a clip attached. In the artefact whose entire job was proving that this repository does not invent evidence.

Found in eleven minutes by a reader who had never seen the work. Fixed, and both the finding and the fix are in the repository.

An unreviewed artefact and a reviewed one are different objects, and the difference is never silent. That rule is in canon, and this is what it bought.

THE PROBLEM

The firehose, and the six events

Detection is table stakes and increasingly commodity at the edge. CameraMatics already detects fatigue, distraction, phone use, tailgating and vulnerable road users. What is missing is deciding which of those events deserves a person — and proving the conversation changed anything.

The manager logs in, sees hundreds, reviews six, and stops logging in. That is not a training problem. It is the product working exactly as built.

So the unit of work is not an event. It is a weekly review session that starts full and ends empty.

What this rests on

Derived from 120 words of brief and public product pages. No insider data.

A-02

Event volume, 0.5–3 per vehicle per day. If it is an order of magnitude lower, this is a detection-breadth product and the brief gets rewritten.

A-13

No triage exists today. If it does, the requirements survive and the story collapses.

A-20

Collision cost concentrates in a minority of drivers. The premise of the commercial case.

Grade and route. Nothing is discarded.

Critical

Interrupt. Clip uploads unconditionally, and an email carrying almost nothing.

High

This week's review set. A person watches the clip and decides.

Medium

Passed to the driver as a group. No conversation.

Low

Recorded. Contributes to the record and the trend. Nobody is asked to look.

This started as a suppression model

Most events are noise, so filter them out and show the rest. That was the agent's design and it was the wrong frame.

Under it, the critical failure is “we hid something real”, which nobody can check because the evidence is what was hidden.

Grading moves the failure to “we graded a conversation down to a notification” — which is checkable, because the event is still there.

Severity decides how much attention. A separate axis decides what kind of event it was — risk, not their doing, or something they got right.

The device decides what to send, the cloud what to show

In-cab device

Detect the v1 triggers on-device in real time. Attach every signal the device has, whichever one fired. Buffer continuously so a clip can be fetched days later.

THE TRADE-OFF

Cellular is contractually capped, so severity gates upload — metadata always, clips when they earn it. Anything collision-grade goes now; that is the one case where cost does not get a vote.

Cloud platform

Ingest idempotently from queue-and-forward devices. Group related detections into one event. Grade, route, and record why. Orchestrate video and never lie about its state.

THE TRADE-OFF

Grading lives here, not in firmware, because these rules will change weekly in the first quarter and a firmware release is the wrong unit of iteration for a number we are guessing.

Web portal

A finite ranked set that can be finished. Honest footage states. A coaching action cheap enough that logging it beats not logging it.

THE TRADE-OFF

It ranks nothing itself and offers no configuration. Deciding what deserves attention is our job, not a slider we hand to the customer.

And the portal decides nothing — it shows a short list honestly. Full outline, constraints and open dependencies in the repository.

The manager's week, and everything underneath it

	1 Events arrive and get graded Before the manager sees anything	2 Start the week What needs me, and what happened to last week	3 Clear the set Move through it without it costing an afternoon	4 Understand the event Work out what actually happened	5 Decide and act Coach, recognise, record or dismiss	6 See whether it worked Answer "is this working" honestly
v1 first shippable 14 stories	ADR-0012 · platform Four triggers, every signal The trigger says an event exists; it does not decide what evidence it carries ADR-0002 · platform Severity-gated video upload Metadata always, clips when they earn it ADR-0007 · platform Four bands, each with an action Nothing is discarded — the band decides what happens next ADR-0010 · platform Risk, neutral or credit What kind of event, separately from how much attention	US-001 A weekly set that can be finished must · M US-002 Critical events bypass the rhythm must · S US-003 The set closes, visibly must · M	US-004 Dismiss a group with a reason should · S — in v1 because R-11 depends on it US-009 Dismiss one, with a reason must · S	US-005 Watch the clip, either side must · M US-006 See why it was graded and classified must · M US-007 Footage states are honest must · M US-008 Resolve who was driving must · M — gates M-03	US-010 Log a coaching conversation must · M — gated on watching the clip US-011 Capture what the driver said must · XS US-012 Confirm recognition must · M — praise the good US-013 Send a medium group to the driver should · S — in v1 so every band has an action	US-014 A driver record showing both sides must · M — coaching, disputes and recognition on one page

Backbone left to right, stories underneath. This is the v1 row — a complete walk across the whole backbone, which is what “the unit of value is the loop” looks like drawn rather than argued. The v1.1 and later rows, and a link from every card to its requirement, are in the repository.

A set that starts full and can be finished

Northgate Logistics42 vehicles · safety review, week of 8–14 September 2026

Sam Okafor

1 critical event needs review

This week's review set

11 to go · ranked by severity, highest first

11

Critical and high — need you this week

38

Medium — driver informed, no conversation

112

Low — recorded only

Critical

Risk — proposed

Hard braking

30 mph zone, Ashby Road · Monday 8 September, 07:42

Marek Nowak · YJ19 KTX

Proposed as risk: the vehicle ahead was stationary and in view for 4 seconds before the brake.

High

Credit — proposed

Hard braking

30 mph zone, Mill Lane · Tuesday 9 September, 14:06

Priya Raman · LM70 VDC

Proposed as credit: a cyclist entered from the left 0.8 s before the brake, and there was no contact.

High

Neutral — proposed

Hard braking

30 mph zone, Farnham Road · Saturday 13 September, 08:05

Joe Mensah · KX67 TRB

Proposed as neutral: a vehicle crossed in from the left lane 1.1 s before the brake.

Working prototype, not a picture — the queue actually empties, and the group of five on one trip is one judgement. Link on slide 3.

Ranked, and finite

A feed is never done. A queue can be cleared, and that is the only state that produces a habit.

Nothing is hidden

The band counts sit on the same screen. 150 events were graded and handled without the manager this week.

Critical breaks the rhythm

It does not wait for the weekly set and it is not closed with it.

Last week closed, not carried

A bad week does not become a backlog nobody will ever face.

Every event carries every signal, whichever one set it off

[Review set](#)

Hard braking. 40 mph zone, Coldharbour Lane.

High

Risk — proposed

[Dee Fitzgerald](#) · BK21 OHP · Tuesday 9 September, 09:14 · E-121

Road-facing camera · playing

-6s

start -3s

+6s

Road-facing

Driver-facing

Everything the device recorded

Speed at the event 38 mph in a 40 mph limit	Force 0.71 g, 0.9 s · triggered	Where Coldharbour Lane, Rugby	When 09:14, Tuesday	Conditions Wipers off, lights off
Forward camera Vehicle ahead, in view 9 s	Driver-facing camera No phone detected			

The trigger decides that an event exists. It does not decide what evidence the event carries.

A hard brake and a phone in hand arrive with the same packet: speed against the limit, the force trace, where, when, wipers and lights, what the forward camera saw, what the driver-facing camera saw.

That is what lets the same event be read as risk or as credit without building a second detector — and it is why the grading reason can be written in a sentence a driver would accept.

Where a device generation cannot flag objects ahead, the panel says so rather than leaving a blank.

Decided on day four, against an earlier decision that had cut it — and two acceptance criteria were already quietly depending on it.

The system proposes. The manager decides. Always.

What kind of event was this, and what happens next?

Proposed as risk: the vehicle ahead was in view for 9 seconds before the deceleration.

Risk — this was their handling
Opens coaching or dismissal

Neutral — not their doing
Records it on Dee's side and closes

Credit — they avoided harm
Records the recognition above

Nothing is preselected

The proposal is named in the panel above, not marked on a button. All three cost the same click.

Cost tracks consequence

Recording something as not the driver's doing needs no clip. Crediting or coaching them does, because those reach a driver.

“The system said so” is not an account

Neutral is the dangerous one: it removes a conversation, and nobody goes looking for conversations that never happened.

Recognition was confirmed by a human from the start. Extending that to every classification came out of building the screen — the requirement had said it and the design had only done it once.

A record with two sides, and a number that admits its limits

[Review set](#)

Marek Nowak
What happened, what was said, and what they got right — on one page.

This week

2 events in the critical and high bands. 0 recorded as not their doing.

Coaching conversations

Phone in hand · 12 August 2026
Talked about. Watched the clip together. The call was from the depot and Marek thought answering was quicker than pulling over.
Agreed. Depot calls go to voicemail; he pulls in at the next layby and calls back.
What Marek said. Said he had not realised it had been that long.
Clip attached · recorded by Sam Okafor

Two questions, not one box

What you talked about, and what you agreed. One box accepts “spoke to driver” — the prototype proved it, so the requirement changed.

The driver's own words

Optional, and recorded verbatim. A dispute is marked and kept beside the event rather than filed somewhere else.

“Too early to say”

Shown for four weeks after a conversation. No figure at all, because one driver over four weeks cannot separate coaching from a quiet month.

Coaching is unavailable until the clip has been played. An unfair conversation costs more trust than ten missed events.

What “implementation-ready” means here

US-010 — Record a coaching conversation from the clip

As a fleet manager I want to log the conversation in one step while I am looking at the footage, so that recording it costs less than not recording it.

- Given the manager has not played the clip, when they attempt to log a conversation, then the action is unavailable and reads “Watch the clip before recording a conversation”.
- Given the manager logs a conversation, then three fields are required: the behaviour, what you talked about, and what you agreed.
- Given any of the three is empty, when they attempt to save, then the action is unavailable. “Spoke to driver” is explicitly not an acceptable record — it is the tick-box this requirement exists to prevent.
- Given a record is saved, then it contains the clip reference, what was discussed and agreed, and the driver's response — enough for a third party to see what happened rather than that something happened.
- Given a record is created, then it attaches to the driver, the event and the footage, and is retrievable as one artefact the manager could show the driver.

Splitting one free-text box into two is a change the prototype forced. Sincerity is not something a field can validate — but a phrase that describes neither what was said nor what was agreed cannot fill both boxes. Fifteen stories are written to this standard; the copy quoted inside an acceptance criterion is the copy that ships.

The unit of value is the loop, so v1 is a thin whole one

What ships first

14 of 15 stories. A manager with a queue and no way to act on it has a better dashboard; a manager with a coaching form and no triage has the firehose with extra steps. Neither half is worth shipping alone.

What waits, and why

Only the fleet roll-up. A trend drawn over four weeks of a low base rate can only mislead, and one misleading number makes every other number on the page suspect.

What capacity buys

Three tracks run concurrently, so people convert into calendar time up to about three teams. What they do not buy is firmware lead time — months, the critical path, and not ours to schedule.

The useful consequence of the dependency chain

The portal does not wait for the edge. It can be built against a stubbed event source from day one, which makes the phases concurrent rather than serial — and it is the only track that produces something to put in front of a fleet manager early enough to change what gets built. I would start there despite it being last in the data flow.

Phase boundaries are conditions, not dates. A boundary that is a date is a guess wearing a calendar.

How fast can you know anything?

The brief names two readers. The operator owns the outcome and signs the renewal; the manager does the work. They ask different questions on different clocks.

Week 1	Are they opening it, and do they trust what is in it	Us — says the machine runs, nothing about safety
Week 4	Risky events per 1,000 miles. Coaching coverage of the drivers generating the risk	The operator — the first honest answer to “is it working”
Week 8–12	Coached-driver repeat rate. Share of drivers in the high-risk band	The operator — attributable to coaching, and slower for it
6–12 months	At-fault collisions, and claim cost	The operator — what they are actually buying
Continuous	Of at-fault collisions, how many had a warning in front of them	Us — whether the queue points at the right population at all

The middle of that ladder is the part worth defending: collisions cannot be read inside a quarter at one fleet, and the repeat rate needs attribution and a control cohort we may not be able to build ethically. The events themselves need neither.

Two gates, and three risks this design created

The gates — these end it rather than cost it money

A-02 Event volume

Breaks the product. If volume is an order of magnitude lower, attention is not the scarce resource and this is a detection-breadth product. Thirty days of telemetry settles it.

A-13 No triage today

Breaks the story, not the requirements. If they already rank and never marketed it, a queue is still a queue — but the framing collapses in front of the person who built the thing. One question answers it.

Risks this design created

Mis-grading

Grading instead of suppressing moved the critical failure rather than removing it: an event that deserved a conversation, sent as a notification.

Unearned praise

Recognising the wrong thing is worse than silence. The drivers who did it properly notice, and it takes the coaching side down with it.

Reporting non-use

Closing the weekly set is right. Counting the closures and displaying the number is a receipt, not a solution.

I would rather be wrong about the second. Being wrong about it costs a conversation; being wrong about the first costs the premise.

A design that adds no new risks has not made any decisions. Two of these share one mitigation, which is why it needed an owner.

Nothing here measures the events that never happened. Of at-fault collisions, how many had a warning in front of them — that is the one check that can say the queue is pointed at the wrong direction.

Six questions I would not commit engineering without

- | | | | |
|---|-------------|---|-------------|
| 1 Does the portal already triage? | COMMERCIAL | 2 What event volume does firmware actually produce? | ENGINEERING |
| <p>The framing of this whole submission. Asked out loud, first, before defending anything.</p> | | <p>Whether this is a triage product at all. If it is low, we stop and I rewrite the brief.</p> | |
| 3 Which device generations can run the inference? | ENGINEERING | 4 How is driver identity established, and for what share? | ENGINEERING |
| <p>Whether v1 is a software release or a hardware programme — and whether recognition ships in v1.</p> | | <p>Assumed solved for this exercise. If it is not, v1 splits: review ships, coaching waits.</p> | |
| 5 What is the lawful basis per market? | LEGAL | 6 What does a manager's week actually look like? | CUSTOMER |
| <p>Which markets can have footage-based coaching at all. One that cannot needs a different product.</p> | | <p>What we are entitled to claim we replace, and the rhythm the set is built around.</p> | |

A blocking question whose answer only moves a date was not blocking. These four change the plan; the other two change what we are allowed to claim.

Where each part of the brief is answered

1 Product brief / business case

Slides 2, 8 · brain/business-case.md, jtbd.md

2 Solution outline across the stack

Slide 10 · brain/stack-outline.md

3 Detailed portal requirements

Slides 11–16 · 15 stories, 4 epics, working prototype

4 Prioritisation and phasing

Slides 11, 17 · brain/phasing.md, storymap

5 Risks, assumptions, open questions

Slides 19, 20 · 25 assumptions, 13 risks, 10 questions

Working with AI

Slides 4–7 · brain/judgement-log.md, reviews/

If the volume assumption is wrong, so is the submission. Thirty days of telemetry settles it, and I would want that before committing engineering.

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github.com/dannydaley76/
cameramatics-product-brain